



GATE CS / IT 2027

Computer Organization & Architecture

Quick Revision Sheet

Complete GATE syllabus coverage — All topics and subtopics

Instruction Set Architecture

Addressing Modes

- Immediate: operand IS the value
- Register: value in register
- Direct: EA = address field
- Indirect: EA = Memory[address field]
- Register Indirect: EA = content of register
- Indexed: EA = Base + Index register
- Relative: EA = PC + offset

RISC vs CISC

- RISC: fixed length, load-store, many registers, pipelining friendly
- CISC: variable length, complex ops, fewer registers
- RISC: MIPS, ARM, SPARC | CISC: x86, x64

Pipelining

Performance & Hazards

- Ideal CPI = 1 with k-stage pipeline
- Speedup = $n \cdot k / (k + n - 1)$ for n instructions
- CPI = 1 + stall cycles per instruction
- Structural Hazard: resource conflict -> stall/duplicate
- Data Hazard RAW: forwarding or stall
- Control Hazard: branch -> predict or delayed branch
- Forwarding: bypass result to next stage without writing to reg

Memory Hierarchy

Cache

- Direct Mapped: block maps to exactly one cache line
- Set Associative: k-way means k possible locations
- Fully Associative: any block can go anywhere
- Write Through: update cache AND memory
- Write Back: update only cache, write to memory on eviction
- AMAT = Hit time + Miss rate * Miss penalty

Virtual Memory

- Page Table: VPN to PFN mapping
- TLB is a cache for page table lookups
- Page Fault: OS loads page from disk
- Thrashing: excessive page faults, working set model

Number Representation & I/O

IEEE 754

- Single precision: 1 sign + 8 exp + 23 mantissa
- Double precision: 1 sign + 11 exp + 52 mantissa
- Bias: 127 (single), 1023 (double)
- Value = $(-1)^S * 1.\text{mantissa} * 2^{(\text{exp}-\text{bias})}$

I/O

- Polling: CPU checks device status in loop
- Interrupt: device notifies CPU when ready
- DMA: device transfers data directly to memory
- Memory-mapped I/O vs Isolated I/O